PANTIUM

(Pantoprazole Sodium Tablets 20mg and 40mg)

Pantoprazole is a substituted benzimidazole. It is a specific proton pump inhibitor. It is useful for the treatment of duodenal ulcer, gastric ulcer, and moderate to severe reflux oesophagitis.

DOSAGE FORM Tablet

DESCRIPTION:

PANTIUM-20

Yellow coloured, round, biconvex, beveled edges, enteric coated tablets.

PANTIUM-40

Yellow coloured, round biconvex, beveled edges enteric coated tablets.

PHARMACODYNAMICS:

Mechanism of Action

Pantoprazole inhibits proton pump in a dose proportionate manner. It inhibits H+ /K+- ATPase enzyme which is responsible for acid secretion in the parietal cells of the stomach, in the acidic environment of the parietal cells, pantoprazole gets converted into the active form, a cyclic sulphenamide, which then binds to the H-/K+- ATPase, thus inhibiting the proton pump. This results in potent and long lasting suppression of the basal and the stimulated gastric acid secretion. Since pantoprazole acts distally to the receptor level, it can inhibit gastric acid secretion irrespective of the nature of the stimulus (acetylcholine, gastrin, histamine). Pantoprazole exerts its full therapeutic effect in a strongly acidic environment ($pH<3$) remaining mostly inactive at higher pH values. Pantoprazole has the same effect whether administered orally or intravenously.

During treatment with antisecretory medicinal products, serum gastrin increases in responses to the decreased acid secretion. Also CgA increases due to decreased gastric acidity. The increased CgA level may interfere with investigations for neuroendocrine tumours.

Available published evidence suggests that proton pump inhibitors should be discontinued between 5 days and 2 weeks prior to CgA measurements. This is to allow CgA levels that might be spuriously elevated PPI treatment to return to reference range.

PHARMACOKINETICS:

Absorption:
Pantoprazole is rapidly absorbed after oral administration and maximum plasma concentration is achieved even after one single 40mg oral dose. Cmax values range from 2-3 microgram/ml after about 2.5 hours post-administration and remain constant after multiple administrations. The absolute Bioavailability of Pantoprazole is about 77% and concomitant administration of food and antacids has no influence on AUC, Cmax, and thus Bioavailability of Pantoprazole.

Distribution:
Pantoprazole is 98% bound to plasma proteins. Volume of distribution of Pantoprazole is about 0.15L/kg.

Metabolism:
Pantoprazole is exclusively metabolized in the liver. The main metabolite both in the plasma and urine is dimethyl Pantoprazole which is conjugated with Sulphate. The half-life of the main metabolite (about 1.5 hours) is not much longer than that of Pantoprazole.

Excretion:
About 80% of the metabolites are eliminated via renal route; the rest are excreted in feces. Terminal half-life is about 1 hour and clearance is about 0.1 L/h/kg. Pharmacokinetics of Pantoprazole does not vary after single or repeated administration and the plasma kinetics are linear after both oral and intravenous administration.

INDICATIONS:
It is indicated in the treatment of
- Duodenal ulcer
- Gastric ulcer
- In combination with two appropriate antibiotics (see recommended dosage) for the eradication of Helicobacter pylori in patients with peptic ulcers with the objective of reducing the recurrence of duodenal and gastric ulcers caused by this microorganism.
- Moderate and severe cases of inflammation of the esophagus (reflux esophagitis).
- Zollinger-Ellison-Syndrome and other pathological hypersecretory conditions.

RECOMMENDED DOSAGE, DOSAGE SCHEDULE AND ROUTE OF ADMINISTRATION:
<u>Recommended Dosage for Pantium 20mg</u>
Dosage and method of administration
Unless otherwise prescribed, the following oral doses apply. Please follow these instructions otherwise there is the risk that the drug may not act properly.

Adults and adolescents 12 years of age and above:
Mild reflux disease and associated symptoms (e.g. heartburn, acid regurgitation, pain on swallowing)
The recommended oral dosage is one Pantoprazole-20 mg per day (1 tablet of Pantium 20mg tablets).

Long-term management and prevention of relapse in reflux oesophagitis

For long-term management, a maintenance dose of one Pantoprazole 20 mg per day is recommended, increasing to 40 mg pantoprazole per day if a relapse occurs. Pantium 40 mg is available for this case. After healing of the relapse the dosage can be reduced again to 20 mg pantoprazole.

Adults:
Prevention of gastroduodenal ulcers induced by non-selective non-steroidal anti-inflammatory drugs (NSAIDs) in patients at risk with a need for continuous NSAID treatment.
The recommended oral dosage is one Pantoprazole 20 mg per day (1 tablet of Pantium 20mg).

Children below 12 years of age:
Pantium 20mg is not recommended for use in children below 12 years of age due to limited data in this age group.

<u>Recommended Dosage for Pantium 40mg:</u>
The following information applies unless Pantium has been otherwise prescribed by your doctor. Please follow these instructions, as otherwise Pantium may not have the desired effect.
In cases of duodenal ulcer or gastric ulcer in which infection with Helicobacter pylori has been confirmed, the microorganism should be eradicated by combination treatment.

Depending on the resistance pattern, the following combinations are recommended:

a) 2 x 1 Pantium 40mg enteric coated tablet/day
\+ 2 x 1000 mg amoxicillin/day
\+ 2 x 500 mg clarithromycin/day

b) 2 x 1 Pantium 40mg enteric coated tablet/day
\+ 2 x 500 mg metronidazole/day
\+ 2 x 500 mg clarithromycin/day

c) 2 x 1 Pantium 40mg enteric coated tablet /day
\+ 2 x 1000 mg amoxicillin/day
\+ 2 x 500 mg metronidazole/day

If combination therapy is not an option, e.g. if the patient has tested negative for Helicobacter pylori, the following dosage guidelines apply for Pantium monotherapy:

For duodenal ulcer, gastric ulcer, and reflux esophagitis:
Generally, Pantium 40mg enteric coated tablet daily.
In individual cases the dose may be doubled (pantoprazole 80mg per day), particularly when there has been no response to other medicines.

Prior to treatment of gastric ulcer, the possibility of malignancy should be excluded as treatment with pantoprazole may alleviate the symptoms of malignant ulcers and can thus delay diagnosis.

Patients with impaired liver function:

In patients with severe liver impairment the dose has to be reduced to 40 mg pantoprazole every other day.

Furthermore, in these patients the liver enzymes should be monitored during pantoprazole therapy. In the case of a rise of the liver enzymes pantoprazole should be discontinued.

Elderly patients and Patients with impaired renal function:
The daily dose of 40 mg pantoprazole should not be exceeded in elderly patients or in patients with impaired kidney function. An exception is combination therapy for eradication of Helicobacter pylori, where also elderly patients should receive the appropriate pantoprazole dose (2 x 40 mg per day) during the 1 -week treatment period.

For the long-term management of Zollinger-Ellison-Syndrome and other pathological hypersecretory conditions:
The recommended daily dose at the beginning of the treatment is pantoprazole 80 mg. Thereafter, the dosage can be titrated up or down as needed using measurements of gastric acid secretion to guide. With doses above 80 mg daily, the dose should be divided and given twice daily. A temporary increase of the dosage above 160 mg pantoprazole is possible but should not be applied longer than required for adequate acid control.

Type and duration of treatment
Combination therapy for eradication of Helicobacter pylori infection usually lasts 7 days and can be extended to a maximum of 2 weeks, if after this time further treatment of pantoprazole is indicated to ensure that ulcer heals completely, the dose recommendations for gastric and duodenal ulcers must be observed.

In the majority of cases, a duodenal ulcer heals completely within 2 weeks. If a two-week treatment period is not sufficient, healing will be achieved in almost all cases within further 2 weeks. Gastric ulcers and reflux esophagitis usually require a 4 week course of treatment. If this should be inadequate, healing will in most cases be achieved within a further 4 weeks. Treatment should not exceed 8 weeks as experience with long-term use is limited.
Treatment duration in Zollinger-Ellison syndrome and other pathological hypersecretory conditions is not limited and should be adapted according to clinical needs.

Instructions for use / handling
The enteric coated tablet of Pantium must not be crushed or chewed and must be swallowed whole with water 1 hour before breakfast.
In combination therapy for eradication of Helicobacter pylori infection, the other 40 mg pantoprazole should be taken before the evening meal

Children: There is no data regarding the use of Pantoprazole in children. Hence, the use of Pantoprazole in children is not recommended.

Route of administration:
Pantium-20 & Pantium-40 are tablets for oral use.

CONTRAINDICATIONS:
Pantoprazole tablets must not be used in combination treatment for eradication of Helicobacter pylori in patients with moderate to severe liver or kidney function disturbances since currently no clinical data are available on the efficacy and safety of pantoprazole in combination treatment of these patients.
Pantoprazole tablets should generally not be used in cases of known hypersensitivity to one of the constituents of pantoprazole tablets or of the combination partners. Pantoprazole, like other PPIs, should not be co-administered with atazanavir (see section interactions).

SIDE EFFECTS / ADVERSE REACTIONS:
See table below:

Frequency **Organ System**	**Common (>1/100, <1/10)**	**Uncommon (>1/1000, <1/100)**	**rare**	**Very rare (<1/10000, incl. isolated reports)**
Blood and lymphatic system				Leucopenia: Thrombocytopenia
Gastrointestinal disorders	Upper abdominal pain: diarrhea; constipation; flatulence ; Fundic gland polyps (benign)	Nausea / vomiting	Dry mouth	Microscopic colitis (Frequency not known)
General disorders and administration site conditions				Peripheral edema
Hepatobiliary disorders				Severe hepato cellular damage leading to jaundice with or without hepatic failure.
Immune system disorders				Anaphylactic reactions including anaphylactic shock with its typical symptoms such as dizziness, rapid pulse or profuse sweating.
Investigations				Increased liver enzymes

				(transaminases, ɣ-GT); elevated triglycerides; increased body temperature.
Musculoskeletal, connective tissue disorders		Fracture of the hip, wrist or spine	Arthralgia	Myalgia
Nervous system disorders	Headache	Dizziness; disturbances in vision (blurred vision)		
Psychiatric disorders			Depression, Hallucination, Disorientation and confusion, Especially in Pre-disposed patients, as well as the aggravation of these symptoms in case of pre-existence .	
Renal and urinary disorders				Interstitial nephritis
Skin and subcutaneous tissue disorders		Allergic reactions such as puritus and skin rash		Urticaria; swelling of the skin or mucous membranes (angioedema) severe reactions of the skin and mucous membranes often associated with blistering, with target lesions (Stevens-Johnson syndrome, erythema multiforme, scalded skin syndrome (Lyell's

				syndrome)); increased sensitivity to light (photosensitivity); Subacute cutaneous lupus erythematosus (Frequency not known)
Metabolism and nutritional disorders				Hypomagnesaemia (Frequency not known); Vitamin B12 deficiency
Infections & infestations		Clostridium difficile associated diarrhea		

Effects on the ability to drive and use machines:
Pantoprazole has no effect on the ability to drive and use machines.

PRECAUTIONS / WARNINGS:
Symptomatic response to therapy with Pantoprazole does not preclude the presence of gastric malignancy. In rodents Pantoprazole is carcinogenic and caused rare types of gastrointestinal tumors. The relevance of these animal findings to humans is unknown. No dosage adjustment is necessary in patients with mild or moderate hepatic impairment. The pharmacokinetics of Pantoprazole has not been well characterized in patients with severe hepatic impairment. Therefore, the potential for modest drug accumulation (< 21 %) when dosed once daily needs to be weighed against the potential for reduced acid control when dosed every other day in these patients.

Regular Surveillance
Patients on proton pump inhibitor treatment (particularly those treated for long term) should be kept under regular surveillance.

Subacute Cutaneous Lupus Erythematosus (SCLE)
Proton pump inhibitors are associated with very infrequent cases of subacute cutaneous lupus erythematosus (SCLE). If lesions occur, especially in sunexposed areas of the skin, and if accompanied by arthralgia, the patient should seek medical help promptly and the health care professional should consider stopping Pantoprazole. SCLE after previous treatment with a proton pump inhibitor may increase the risk of SCLE with other proton pump inhibitors.

Hypomagnesaemia
Severe hypomagnesaemia has been reported in patients treated with PPI like Pantoprazole for at least three months, and in most cases for a year. Serious manifestations of hypomagnesaemia such as fatigue, tetany, delirium, convulsions, dizziness and ventricular arrhythmia can occur but they may begin insidiously and be overlooked. In most affected patients, hypomagnesaemia improved after magnesium replacement and discontinuation of the PPI.

For patients expected to be on prolonged treatment or who take PPI with digoxin or drugs that may cause hypomagnesaemia (e.g., diuretics),health care professionals should consider measuring magnesium levels before starting PPI treatment and periodically during treatment.

Fracture
Proton pump inhibitors, especially if used in high doses and over long durations (>1 year), may modestly increase the risk of hip, wrist and spine fracture, predominantly in the elderly or in presence of other recognized risk factors. Observational studies suggest that proton pump inhibitors may increase the overall risk of fracture by 10–40%. Some of this increase may be due to other risk factors. Patients at risk of osteoporosis should receive care according to current clinical guidelines and they should have an adequate intake of vitamin D and calcium.

Clostridium Difficile Diarrhea
Published observational studies suggest that PPI therapy may be associated with an increased risk of Clostridium difficile associated diarrhea, especially in hospitalized patients. This diagnosis should be considered for diarrhea that does not improve. Patients should use the lowest dose and shortest duration of PPI therapy appropriate to the condition being treated.

Vitamin B12 Deficiency
Daily treatment with any acid-suppressing medications over a long period of time (e.g., longer than 3 years) may lead to malabsorption of cyanocobalamin (vitamin B12) caused by hypo- or achlorhydria. Rare reports of cyanocobalamin deficiency occurring with acid-suppressing therapy have been reported in the literature. This diagnosis should be considered if clinical symptoms consistent with cyanocobalamin deficiency are observed.

Increased Chromogranin A(CgA) level may interfere with investigations for neuroendocrine tumors. If the patient(s) are due to have a test on Chromogranin A level, Pantoprazole treatment should be stopped for at least 5 days before CgA measurements to avoid this interference (see section Pharmacodynamics). If CgA and gastrin levels have not returned to reference range after initial measurement, measurements should be repeated 14 days after cessation of proton pump inhibitor treatment.

POSTMARKETING EXPERIENCE
Risk of microscopic colitis.

PREGNANCY AND LACTATION:
There is no data regarding the use of Pantoprazole during pregnancy and breast feeding. The drug should be used only if benefits outweigh the risks.

INTERACTION:
No clinically significant drug interactions have been reported with Pantoprazole. Studies in humans reveal no interaction of Pantoprazole with cytochrome P450-system of the liver. Concomitant administration of Pantoprazole with either antipyrine, diazepam, phenytoin, nifedipine, theophylline, digoxin or oral contraceptives does not result in inhibition of metabolism. Also simultaneous administration of Pantoprazole with warfarin does not influence warfarin's effect on coagulation factors.

Interaction between Pantoprazole and Atazanavir: It has been shown that co-administration of Atazanavir 300mg/ Ritonavir 100mg with Omeprazole (40mg once daily) or Atazanavir 400mg with Lansoprazole (60mg single dose) to healthy volunteers resulted in a substantial reduction in the bioavailability of atazanavir. The absorption of atazanavir is pH dependent. Therefore, PPIs, including pantoprazole, should not be co-administered with atazanavir.

SYMPTOMS AND TREATMENT FOR OVERDOSAGE, AND ANTIDOTE (S):
There are no known symptoms of overdosage in man. However, Pantoprazole is very specific in action and no particular problems are anticipated. As Pantoprazole is extensively protein bound, it is not readily dialysable. Apart from symptomatic and supportive treatment, no specific therapeutic recommendation can be made.

STORAGE CONDITIONS:
Store in a cool & dry place at 30°C, protected from light.

SHELF LIFE:
24 months from the date of manufacturing.

PACK SIZE:
Pantium-20 & Pantium-40 (Pantoprazole Sodium Tablets 20 mg and 40mg) are available in blister strips of 10 tablets packed in a printed carton.
Each carton contains 10 such strips. (10 × 10 T).

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